Tables and graphs

1 A diver starts at the surface of the water and starts to descend below the surface at a constant rate. The table below shows the depth of the diver over 4 minutes:

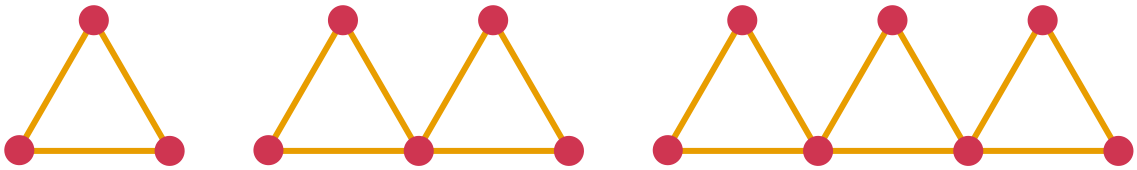
Number of minutes passed (x)	0	1	2	3	4
Depth of diver in metres (y)	0	1.35	2.7	4.05	5.4

a Write an equation that describes the relationship between the number of minutes passed (x) and the depth (y) of the diver.

b At what depth would the diver be after 59 minutes?

c What does the numerical coefficient in the equation from part (a) represent?

2 Matches were used to make the pattern below:



a Complete the table below:

Number of triangles (t)	1	2	3	5	10	20
Number of matches (m)						

b Write an equation that describes the relationship between the number of matches (m) and the number of triangles (t).

c Hence, how many matches are required to make 21 triangles?

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3 A racing car starts the race with 80 litres of fuel. From there, it uses fuel at a rate of 2 litres per minute.

a Complete the table below:

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in tank (y)					

b Write an algebraic relationship linking the number of minutes passed (x) and the amount of fuel left in the tank (y).

c After how many minutes, x, will the car need to refuel (i.e. when there is no fuel left) ?

d Explain why plotting the points from the table in part (a) results in a straight line graph.

4 There are 40 litres of water in a rainwater tank. It rains for a period of 24 hours and during this time, the tank fills up at a rate of 10 litres per hour.

a Complete the table below:

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

b Write an algebraic relationship linking the number of hours passed (x) and the amount of water in the tank (y).

c Sketch the graph of the linear equation from part (b).

5 It starts raining and an empty rainwater tank fills up at a constant rate of 2 litres per hour. By midnight, there are 10 litres of water in a rainwater tank. As it rains, the tank continues to fill up at this rate.

a Complete the table below:

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

b Write an algebraic relationship linking the number of hours passed since midnight (x) and the amount of water in the tank (y).

c How many hours before midnight was the tank empty (i.e. when $y = 0$)? Note that x represents the number of hours passed since midnight, so $-x$ represents x hours before midnight.

d Sketch the graph of the linear equation from part (b).

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6 Consider the table below:

Time in minutes (x)	1	2	3	4	5
Temperature in $^\circ$C (y)	5	9	13	17	21

a By how much is the temperature increasing each minute?

b What would the temperature have been at time 0 minutes?

c Find the algebraic relationship between x and y.

d Sketch the graph of the linear equation from part (c).

7 In a study, scientists found that the more someone sleeps, the quicker their reaction time. The attached table shows the findings:

Number of hours of sleep (x)	0	1	2	3	4	5
Reaction time in seconds (y)	9	8.8	8.6	8.4	8.2	8

a How much does the reaction time change for each extra hour of sleep?

b Write an algebraic equation relating the number of hours of sleep (x) and the reaction time (y).

c Find the reaction time for someone who has slept 1.5 hours.

d Find the number of hours someone sleeps if they have a reaction time of 8.5 seconds.

Word problems

8 Consider the statement: 9 divided by the sum of x and 3 is equal to 6 divided by the difference between x and 3.

i Write an equation for the statement.

ii Solve the equation for x.

9 Eileen tries to guess how many people are at a concert, but she guesses 400 too many. Adam guesses 250 too few. The average of their guesses is 2575. Set up an equation and solve for x, the exact number of people at the concert.

10 Kate and Isabelle do some fundraising for their sporting team. Together they raised $\$600$. If Kate raised $\$272$ more than Isabelle, and Isabelle raised $\$p$:

a Write an equation in terms of p that represents the relationship between the different amounts and solve for p.

b Hence, calculate how much Kate raised.

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11 Caitlin's phone plan allows her to send 300 text messages for free. But if she sends over 300 messages, the cost, C, in dollars of sending x text messages is modelled by:

$$C = 0.1\,(x - 300) + 8$$

Solve for the number of text messages x that have been sent when the total cost is $\$30.10$.

12 To manufacture doors, the manufacturer has a fixed cost of $\$4200$ plus a variable cost of $\$100$ per door. Find n, the number of doors that need to be produced so that the average cost per door is $\$120$.

13 Eileen's sink can be filled in 8 minutes and can subsequently be drained in 16 minutes when it is unplugged. Find the number of minutes, n, that it will it take to fill the sink if it is left unplugged while being filled.

14 Oprah can paint the average house in 15 hours, while Tobias can do it in 10 hours. Find the number of hours, t, that it would take to paint a house if they both worked together.

15 A phone salesperson earned $\$880$ in a particular week during which she sold 22 units. In another week she made $\$925$ and sold 25 units. Let x be the number of units sold and y be the weekly earnings.

a Find the linear equation that models the units-earnings relationship for this salesperson.

b Hence, predict the earnings of the salesperson if she sells 36 units.

16 Sisters, Ursula and Eileen, are training for a triathlon event. Ursula finds that her average cycling speed is 13 kph faster than Eileen's average running speed. Ursula can cycle 46 km in the same time that it takes Eileen to run 23 km.

a If Eileen's running speed is n km per hour, solve for n.

b Find Ursula's average cycling speed.

17 When a new car was released on the market, car journalists, Susana and Irene, were invited to take the car for a test drive. Each took their car into the mountains, where Susana covered a distance of 31 km in the same time that Irene drove 25 km. Susana drove 12 kph faster than Irene.

a If Irene drove at a speed of n kph, solve for n.

b Hence, find Susana's speed.

Geometric applications

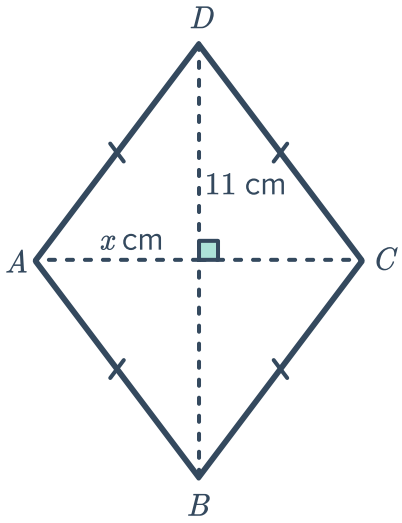
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18 Find the perpendicular height, h, of a parallelogram that has an area of $66\ \text{cm}^2$ and a base of $6\ \text{cm}$. The area of a parallelogram is given by: $A = bh$.

19 Find the base length, b, of a parallelogram that has an area of $228\ \text{mm}^2$ and a perpendicular height of $12\ \text{mm}$. The area of a parallelogram is given by: $A = bh$.

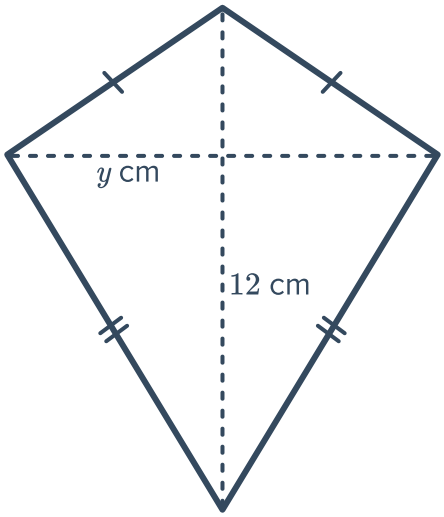
20 Find the missing length, x, if the area of rhombus $ABCD$ has an area of $44\ \text{cm}^2$. The area of a rhombus is given by:

$$A = \frac{1}{2}xy$$



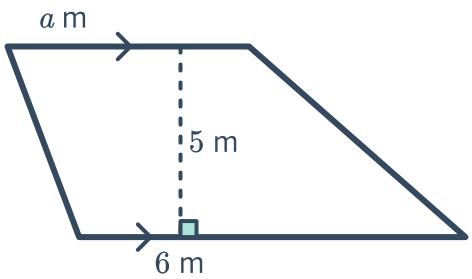
21 Find the missing length, y, if the area of the kite is $48\ \text{cm}^2$. The area of a kite is given by:

$$A = \frac{1}{2}xy$$



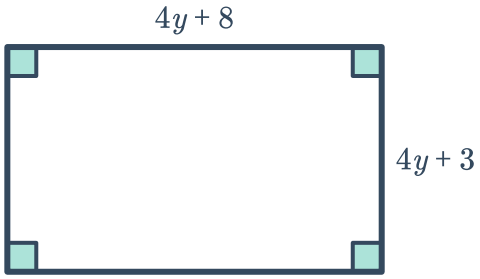
22 Find the missing side length, a, if the area of the trapezium shown is $25\ \text{m}^2$. The area of a trapezium is given by:

$$A = \frac{1}{2}(a+b)\,h$$

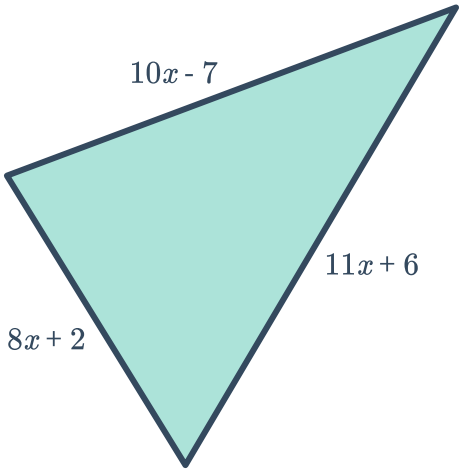


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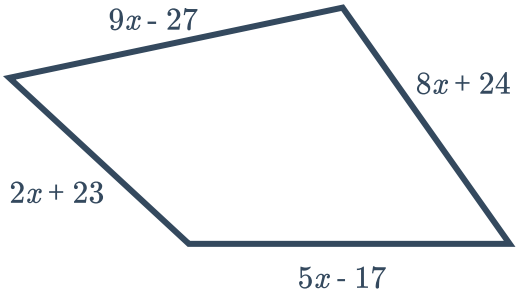
23 The perimeter of the rectangle is $126 + 3y$ cm. Solve for y.



24 The perimeter of the triangle is 146 cm. Solve for x.



25 The perimeter of the quadrilateral is 315 cm Solve for x.



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